

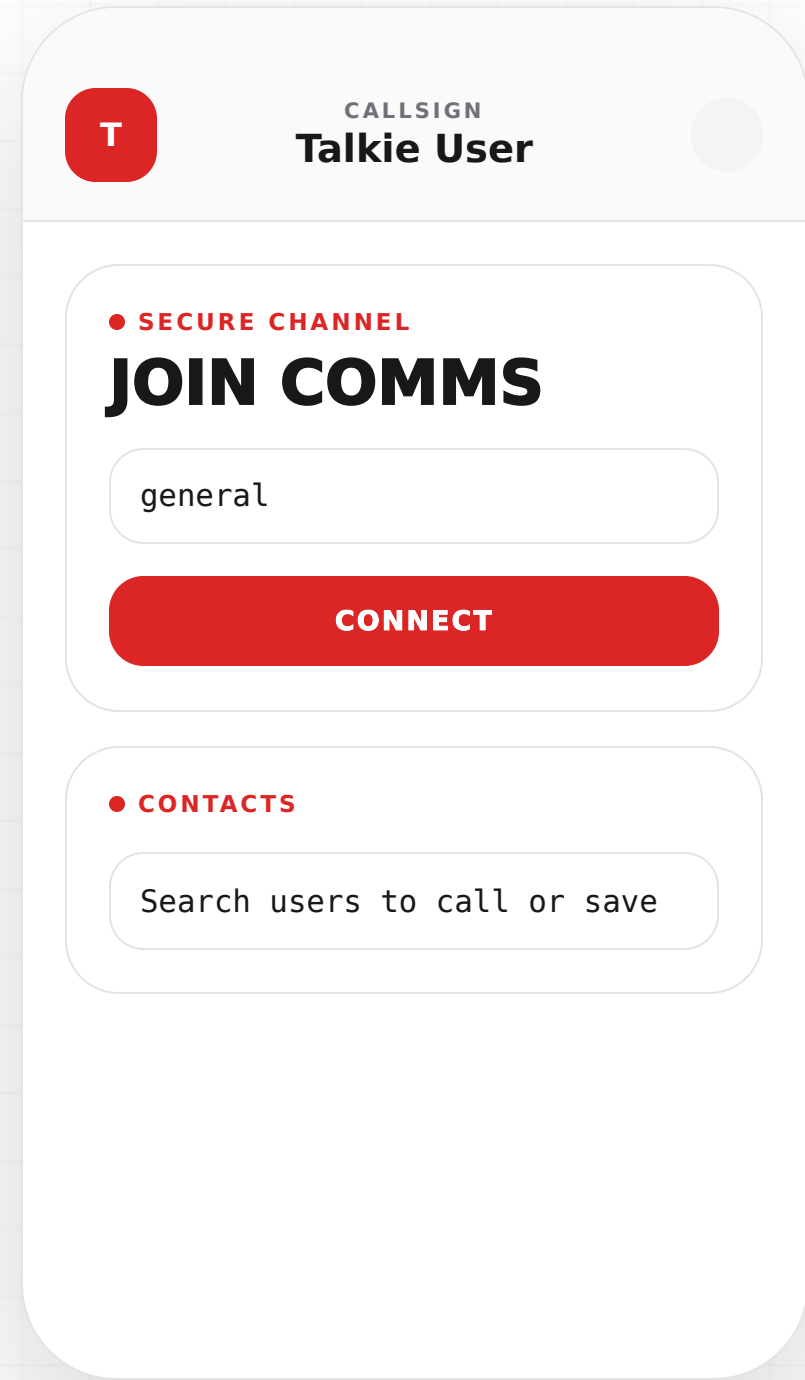


FEATURE OVERVIEW

TALKIE PUSH-TO-TALK APP

A mobile-first web/PWA walkie-talkie experience with secure sign-in, live channels, direct calls, contacts, chat, and browser-local voice replay.

CURRENT PRODUCTION FEATURE SET



IDENTITY AND ACCESS

SECURE ACCOUNTS AND CONTROLLED ACCESS

Clerk authentication

Users sign in and register through Clerk-backed auth screens.

Display names

The app shows first name, username, full name, or email instead of raw user IDs.

Admin controls

Admin-only route support exists for managing users and disabled accounts.

Disabled-user blocking

Token and user search APIs reject disabled accounts before they can join or be called.

Private identities

Direct calls use Clerk user IDs internally while showing readable names in the UI.

Vercel-ready config

The app builds from the web project root and deploys from GitHub main.

PUSH-TO-TALK CORE

LIVE AUDIO CHANNELS POWERED BY LIVEKIT

- **Public channels**

Users can join named live rooms such as general.

- **Push-to-talk mic**

Microphone access is requested lazily when the user presses the talk button.

- **Connection status**

The channel header shows live, connecting, reconnecting, and offline states.

- **PIN-protected channels**

Private rooms are derived from channel name plus PIN.

- **Spacebar support**

Desktop users can hold Space to transmit.

- **Recent channels**

Previously joined channels are remembered locally for quick re-entry.

DIRECT CALLS

ONE-TO-ONE CALLS, CONTACTS, AND INCOMING ALERTS

1

Search

Find users by name or email through Clerk-backed search.

2

Save

Add users to local contacts for one-tap calling later.

3

Call

Create a deterministic private LiveKit room for both users.

4

Alert

Open recipients see an incoming call sheet with Answer and Decline.

5

Connect

The direct channel switches from ringing to live when the peer joins.

Contacts include activity hints based on Clerk account activity. Pending calls are short-lived and protect against self-calling or disabled users.

IN-CHANNEL TOOLS

EVERYTHING NEEDED WHILE A CHANNEL IS LIVE

- **People panel**

See local and remote participants in the room.

- **Per-user volume**

Remote participants can be adjusted independently.

- **Text chat**

Live data-channel chat works inside the same room.

- **Speaking indicators**

Active speakers get visual status and animated VU bars.

- **Mute controls**

Remote participant audio can be muted locally.

- **Unread badges**

Chat and replay tabs show unread counts while users are on another tab.

REPLAY AND FEEDBACK

MISSED AUDIO CAN BE REPLAYED DURING THE SESSION

Browser-local replay

Received audio clips are recorded in the browser while the channel page is open.

Replay tab

Users can listen back to recent clips without leaving the call.

Auto cleanup

Clips are capped and revoked locally; they disappear when leaving the channel.

Incoming beep

Optional tone plays when someone starts speaking.

Vibration

Supported phones can vibrate on incoming speech.

Browser notifications

When enabled, backgrounded tabs can show speaking notifications.

INSTALLABLE MOBILE-FIRST WEB APP

- **Manifest**

Talkie ships a web app manifest with app name, icons, and display mode.

- **iOS install hint**

iPhone users get guidance to add the app to the home screen.

- **Mobile shell**

The UI is designed around a phone-sized viewport with touch-first controls.

- **Service worker**

The app registers a service worker and includes an offline fallback page.

- **Home-screen icons**

Standard, Apple touch, and maskable icons are generated.

- **White/red theme**

Home, channel, and settings screens share the same brand styling.

USER-CONTROLLED AUDIO AND ALERTS

Output volume

Global output level controls remote audio playback.

Beep toggle

Users can turn incoming speech tones on or off.

Vibration toggle

Vibration can be disabled for devices that support it.

Notification toggle

The app requests browser notification permission only when the user enables it.

Permission-aware UI

The settings screen explains unsupported, blocked, and granted notification states.

Local persistence

Settings are saved in browser local storage.

ARCHITECTURE

CURRENT APP STACK

NEXT.JS

REACT

CLERK

LIVEKIT

PWA

VERCEL

LOCAL STORAGE

SERVICE WORKER

- **Frontend**
Next.js app router with client screens for home, channel, settings, auth, and admin.
- **Identity**
Clerk protects the app and supplies user records for names, search, and admin checks.

- **Realtime audio**
LiveKit tokens are issued server-side with room grants for audio and data.
- **Local state**
Recent channels, contacts, PINs, settings, and replay clips are browser-local.